Mathematiktext-Klassifizierung mittels Wikipediadaten

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Motivation

Ich möchte jegliche Technische notizen automatisch klassifierzieren lassen.

Publikumsfrage:

Welchem Mathematik-Themenbereich gehören diese drei Sätze jeweils an?

"... die Zahl 6 ist keine Primzahl, aber die Zahl 5 ..."

"... die Projektionsmatrix P zum Quadrat ist sie selbst, $P^2 = P$ ..."

"... $F[f](\omega) := \int f(t)\mathrm{e}^{-t\omega} dt$ ist auch wieder invertierbar ..."

"... nach dem Satz von Vieta kann man ..."

Demo!

```
Enter text to search and classify (or hit Enter to use default):
I like coordinate transforms in 3d space

Query-embedding dimension: 768, first few values:
[-0.0261, -0.0253, 0.0130, -0.0107, -0.0045, -0.0051, 0.0760, 0.0084, -0.0363, -0.0228 ...]

Closest embedded math wikipedia pages:
 1.  dist = 0.912      Coordinate_systems                  wikipedia.org/wiki/List_of_common_coordinate_transformations
 2.  dist = 0.937      Transformation_(function)           wikipedia.org/wiki/Helmert_transformation
 3.  dist = 0.981      Transformation_(function)           wikipedia.org/wiki/Translation_of_axes
 4.  dist = 0.982      Transformation_(function)           wikipedia.org/wiki/Transformation_matrix
 5.  dist = 1.016      Metric_tensors                      wikipedia.org/wiki/Curvilinear_coordinates
 6.  dist = 1.025      Tensors                             wikipedia.org/wiki/Cartesian_tensor
 7.  dist = 1.033      Transformation_(function)           wikipedia.org/wiki/Rotation_of_axes
 8.  dist = 1.064      Coordinate_systems                  wikipedia.org/wiki/Quadray_coordinates
 9.  dist = 1.065      Projective_geometry                 wikipedia.org/wiki/Homogeneous_coordinates
10.  dist = 1.084      Triangle_geometry                   wikipedia.org/wiki/Trilinear_coordinates
11.  dist = 1.087      Linear_algebra                      wikipedia.org/wiki/Three-dimensional_rotation_operator
12.  dist = 1.091      Clipping_(computer_graphics)        wikipedia.org/wiki/Clip_coordinates
13.  dist = 1.101      3D_computer_graphics                wikipedia.org/wiki/Quaternions_and_spatial_rotation
14.  dist = 1.106      Geometry_stubs                      wikipedia.org/wiki/Planar_projection
15.  dist = 1.115      Geometry_in_computer_vision         wikipedia.org/wiki/Homography_(computer_vision)
16.  dist = 1.123      Geometry_in_computer_vision         wikipedia.org/wiki/Collinearity_equation
17.  dist = 1.124      Geometric_algebra                   wikipedia.org/wiki/Plücker_coordinates
18.  dist = 1.128      Metric_tensors                      wikipedia.org/wiki/Tensors_in_curvilinear_coordinates
19.  dist = 1.129      Geometry_in_computer_vision         wikipedia.org/wiki/Trifocal_tensor
20.  dist = 1.131      Computer_graphics                   wikipedia.org/wiki/Glossary_of_computer_graphics

Wikipedia category counts:
                      |  16.7%   (5 out of 30)  Transformation_(function)
                      |  16.7%   (5 out of 30)  Geometry_in_computer_vision
                      |  13.3%   (4 out of 30)  Coordinate_systems
                      |   6.7%   (2 out of 30)  Metric_tensors
                      |   3.3%   (1 out of 30)  Tensors
                      |   3.3%   (1 out of 30)  Projective_geometry

My custom categorization:
                      |  53.3%   geometry
                      |  10.0%   linear_algebra
                      |   6.7%   analysis
```

```
Enter text to search and classify (or hit Enter to use default):
my friend gave a presentation on the use of Kalman filters in tax time series

Query-embedding dimension: 768, first few values:
[-0.0596, -0.0727, -0.0187, 0.0384, -0.0251, -0.0032, 0.0437, -0.0015, -0.0450, 0.0173 ...]

Closest embedded math wikipedia pages:
 1.  dist = 1.058      Nonlinear_filters                  wikipedia.org/wiki/Kalman_filter
 2.  dist = 1.078      Nonlinear_filters                  wikipedia.org/wiki/Extended_Kalman_filter
 3.  dist = 1.095      Time_series_models                 wikipedia.org/wiki/State-space_representation
 4.  dist = 1.108      Control_theory                     wikipedia.org/wiki/Schmidt–Kalman_filter
 5.  dist = 1.131      Nonlinear_filters                  wikipedia.org/wiki/Switching_Kalman_filter
 6.  dist = 1.155      Nonlinear_time_series_analysis     wikipedia.org/wiki/Nonlinear_autoregressive_exogenous_model
 7.  dist = 1.155      Stochastic_differential_equations  wikipedia.org/wiki/Filtering_problem_(stochastic_processes)
 8.  dist = 1.156      Nonlinear_filters                  wikipedia.org/wiki/Bellman_filter
 9.  dist = 1.170      Time_series_models                 wikipedia.org/wiki/Vector_autoregression
10.  dist = 1.170      Signal_processing                  wikipedia.org/wiki/Autoregressive_model
11.  dist = 1.173      Signal_processing                  wikipedia.org/wiki/Alpha_beta_filter
12.  dist = 1.179      Finnish_statisticians              wikipedia.org/wiki/Pentti_Saikkonen
13.  dist = 1.187      Canadian_mathematicians            wikipedia.org/wiki/Rogemar_Mamon
14.  dist = 1.215      Time_series                        wikipedia.org/wiki/Tracking_signal
15.  dist = 1.216      Time_series_econometricians        wikipedia.org/wiki/Helmut_Lütkepohl
16.  dist = 1.222      Econometric_models                 wikipedia.org/wiki/Econometric_model
17.  dist = 1.230      Multivariate_time_series           wikipedia.org/wiki/Bayesian_vector_autoregression
18.  dist = 1.234      Time_series                        wikipedia.org/wiki/Berlin_procedure
19.  dist = 1.234      Time_series                        wikipedia.org/wiki/Hodrick–Prescott_filter
20.  dist = 1.237      Nonlinear_filters                  wikipedia.org/wiki/Auxiliary_particle_filter

Wikipedia category counts:
                      |  23.3%   (7 out of 30)   Nonlinear_filters
                      |  16.7%   (5 out of 30)   Time_series
                      |   6.7%   (2 out of 30)   Time_series_models
                      |   6.7%   (2 out of 30)   Signal_processing
                      |   3.3%   (1 out of 30)   Control_theory
                      |   3.3%   (1 out of 30)   Nonlinear_time_series_analysis

My custom categorization:
                      |  43.3%   probability_stats
                      |  33.3%   information_crypto
                      |  20.0%   mathematicians
                      |   3.3%   analysis
```

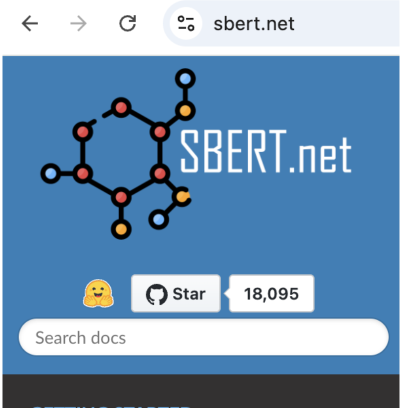
```
Enter text to search and classify (or hit Enter to use default):
In the world of mathematics, matrices are more than just arrays of numbers; they hold the keys to solving complex problems across various fields, from engineering to e
conomics. But what happens when you need to find the inverse of a matrix? This is where things can get tricky. The concept itself might seem daunting at first glance,
but with a little guidance and some handy tools, it becomes much more approachable.

To start off, let's clarify what we mean by 'inverse.' Just like how multiplying a number by its reciprocal gives you one (for example, 2 multiplied by 0.5 equals 1),
in matrix terms, finding an inverse means identifying another matrix that will yield the identity matrix when multiplied together. The identity matrix acts as a neutra
l element in multiplication—think of it as '1' for matrices.
Query-embedding dimension: 768, first few values:
[-0.0271, -0.0440, -0.0145, -0.0105, -0.0310, 0.0328, -0.0019, 0.0132, 0.0102, 0.0133 ...]

Closest embedded math wikipedia pages:
 1.  dist = 0.680       Determinants                          wikipedia.org/wiki/Invertible_matrix
 2.  dist = 0.826       20th-century_Chinese_mathematicians   wikipedia.org/wiki/Guorong_Wang
 3.  dist = 0.905       Matrices                              wikipedia.org/wiki/Integer_matrix
 4.  dist = 0.908       Matrices                              wikipedia.org/wiki/Generalized_inverse
 5.  dist = 0.910       Exchange_algorithms                   wikipedia.org/wiki/Gaussian_elimination
 6.  dist = 0.910       Matrix_theory                         wikipedia.org/wiki/Partial_inverse_of_a_matrix
 7.  dist = 0.918       Numerical_linear_algebra              wikipedia.org/wiki/Moore–Penrose_inverse
 8.  dist = 0.935       Linear_algebra_stubs                  wikipedia.org/wiki/Frobenius_matrix
 9.  dist = 0.963       Linear_algebra_stubs                  wikipedia.org/wiki/Drazin_inverse
10.  dist = 0.977       Numerical_linear_algebra              wikipedia.org/wiki/Inverse_iteration
11.  dist = 0.981       Linear_algebra                        wikipedia.org/wiki/Elementary_matrix
12.  dist = 0.981       Lemmas_in_linear_algebra              wikipedia.org/wiki/Woodbury_matrix_identity
13.  dist = 0.986       Matrix_theory                         wikipedia.org/wiki/Sherman–Morrison_formula
14.  dist = 0.987       Linear_algebra_stubs                  wikipedia.org/wiki/Unimodular_polynomial_matrix
15.  dist = 0.988       Matrices                              wikipedia.org/wiki/Jacket_matrix
16.  dist = 0.994       Computational_number_theory           wikipedia.org/wiki/Itoh–Tsujii_inversion_algorithm
17.  dist = 1.003       Numerical_libraries                   wikipedia.org/wiki/Efficient_Java_Matrix_Library
18.  dist = 1.006       Experimental_mathematics              wikipedia.org/wiki/Inverse_Symbolic_Calculator
19.  dist = 1.020       Matrix_decompositions                 wikipedia.org/wiki/Matrix_decomposition
20.  dist = 1.021       Matrix_multiplication_algorithms      wikipedia.org/wiki/Matrix_multiplication_algorithm

Wikipedia category counts:
                    |  13.3%   (4 out of 30)   Matrices
                    |  13.3%   (4 out of 30)   Linear_algebra_stubs
                    |  10.0%   (3 out of 30)   Numerical_linear_algebra
                    |   6.7%   (2 out of 30)   Determinants
                    |   6.7%   (2 out of 30)   Exchange_algorithms
                    |   6.7%   (2 out of 30)   Matrix_theory

My custom categorization:
                    |  33.3%   analysis
                    |  30.0%   linear_algebra
                    |  13.3%   algos_cs
                    |   6.7%   mathematicians
                    |   3.3%   number_theory
```

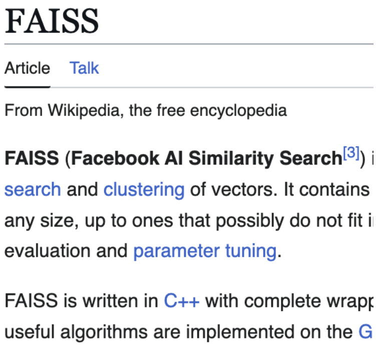
Vortragsinhalt

- Suchprogramm-Komponenten
  - Funktionsweise eines **SentenceTransformers**
- **Wikipedia** und meine Datenerstellung
  - Native Wikipedia-Datenstruktur (“Kategorien”, “Mathe-Seite”)
  - Aufbau meines Datenkorpus: heuristiken, filtern, iteration
  - Analyse-Einsichten, “Fun Findings”
- Aussicht, Pläne

Komponenten

- **Text-Einbettungen via Transformer** (BERT by google, SBERT & TU Darmstadt)
- Ähnlichkeitssuche (FAISS by facebook)
  - bestimmt **argmax für L2-Vektor-Abstand**, heute nicht tiefer besprochen
- Datensatz (EU's "OpenAIRE"/"wikinformetrics" Projekt und offizielle API)





FAISS

Article Talk

From Wikipedia, the free encyclopedia

FAISS (Facebook AI Similarity Search[3]) i
search and clustering of vectors. It contains
any size, up to ones that possibly do not fit i
evaluation and parameter tuning.

FAISS is written in C++ with complete wrapp
useful algorithms are implemented on the G

Publikumsfrage:

Für was steht eigentlich…

Chat

G

P

T

Chat

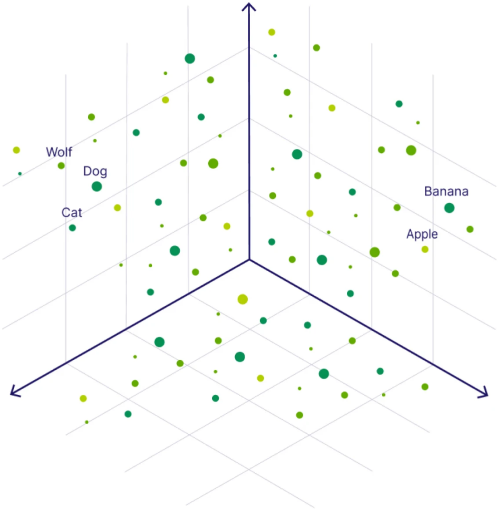
Generative

Pre-trained

Transformer

Startpunkt:

Fixe ursprüngliche Wort-Kodierung in $\mathbb{R}^{786}$, für alle 200k Englishen Wörter.



Startpunkt, links:

Hier representiert jede row von X, links, eine fixe ursprüngliche Wort-Kodierung.

$\mapsto$

Wort	$X_{i,1}$	$X_{i,2}$	$X_{i,3}$	...
Die	-0.41	0.22	-0.67	...
Katze	0.78	-0.33	0.09	...
jagt	0.34	0.47	-0.12	...
hungrig	-0.12	-0.71	0.55	...
die	-0.89	0.15	-0.24	...
Maus	0.56	-0.54	0.31	...

$E(X)_{i,1}$	$E(X)_{i,2}$	$E(X)_{i,3}$	...
-0.47	0.28	-0.62	...
0.72	-0.31	0.19	...
0.10	0.41	-0.09	...
-0.15	-0.78	0.06	...
-0.82	0.14	-0.17	...
0.59	-0.56	0.25	...

“$\mapsto$” representiert die vom eigenen Kontext abhängige Transformer-Abbildung.

Die Einbettung des ganzen Satzes ist (grob) der Mittelwert über die Reihen der Einbettung rechts, also $\mathrm{SatzFeature}_j := \frac{1}{6}\sum_{i=1}^{6} E(X)_{ij}$.

$\mapsto$

Wort
Die
Katze
jagt
hungrig
die
Maus

Beispielhaft: Bedeutung $E(X)$
Die
"hungrige, jagende, ... Katze"
"jagt, läuft"
hungrig
die
"müde, ängstliche, ... Maus"

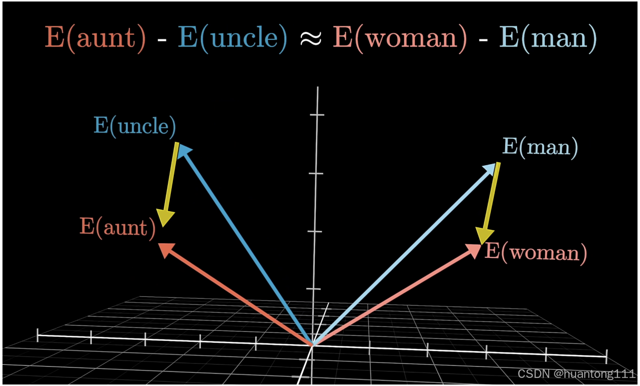
Kontext-Mechanismus:

Jede Zeile von $E(X)$ hängt stark von allen einträgen von X ab.

Training: Im Raum $\mathbb{R}^{786}$ wird mehr kodiert als die 200k Punkte für sich allein!

Rot/Blau: Beispiele für Wort-Einbettungen

Gelb: Richtungen mit **emergent** semantischem Inhalt



Anderes Beispiel. Ein Wort habe ich geändert: **“hungrig” zu “Jerry”**

$$\mapsto$$

Wort	Beispielhaft: Bedeutung $E(X)$
Die	Die
Katze	“wütende, jagende, ... Katze”
jagt	“jagt, läuft”
Jerry	“Comic-Figur ... Maus ... Jerry”
die	die
Maus	“schlaue, braune, ... Maus”

Kontext-Mechanismus:

Jede Zeile von $E(X)$ hängt stark von allen einträgen von X ab.

$\mapsto$

Wort
Apple
is
red

Beispielhaft: Bedeutung $E(X)$
“Fruit Apple”
is
“red color, surface”

$\mapsto$

Wort
Apple
USB
cable

Beispielhaft: Bedeutung $E(X)$
“Apple computer”
“Universal serial bus”
“USB charging cable”

Einbettung von “Apple” wird komplett anderer Vektor, je nach Kontext!

Kennzahlen

- Verwendete Vokabulargröße: **30k**
- Vektorlänge per Wort: **786 dim**
- Max input länge (nativ): **300 Wörter**
- Model weights auf der Platte: **110 Mill. floats / 0.44 GB**

Machbar auf einem Laptop: **3.5k dim**; **20k Wörter**; **16 GB**

(largest open Facebook model: 16.4k dim; 200k Wörter; 500 GB)

- Gezogen Matheseiten: **40k** (700 MB)
- Einbettung (der ersten 300 Wörter ja): auch **2 Stunden**
- Artikel- und Metadata Pullzeit: **3 Std** (30 GB)
- Datenbank-Erstellung zur Analyse: auch **3 Std**

Wikipedia-Daten

„Warum Wikipedia¿‘: offen, groß, verlinkt, Metadaten bereitgestellt.

Englische Wikipedia: ca. **7,1 Mio.** Artikel (gesamt)

Problemstellung: Was ist überhaupt ein “Mathe-Artikel”?

Schlußendlich: Mathe-Artikel-Korpus: 40k Artikel

(i.e., 0.6% der Artikel)

Das Kategorien-Konzept von Wikipedia

Wikipedia hat ein “Kategorien”-Konzept, aber es ist sehr grob:

Auf jede Kategorie kommen circa 3 Artikel! **Über 2 Millionen Kategorien!**

Live erklärung der Kategorien. Startpunkt:

https://en.wikipedia.org/wiki/Benford%27s_law

Für Mathe ist es circa 16, also noch immer 2500 Kategorien.

Allerdings: Man landet schnell außerhalb von Mathe (Geologie, Ökonomie, ...)

```
pageid=22476294 category=Category:Matrix decompositions
pageid=1087836 category=Category:Matrix normal forms
pageid=16230266 category=Category:Domain decomposition methods
pageid=45081487 category=Category:Matrix multiplication algorithms
pageid=31819845 category=Category:Relaxation (iterative methods)
pageid=30994638 category=Category:Fréchet spaces
pageid=7548377 category=Category:Normed spaces
pageid=7430263 category=Category:Sobolev spaces
pageid=30994686 category=Category:Function spaces
pageid=30994559 category=Category:Metric linear spaces
pageid=10973513 category=Category:Spacetime
pageid=25261497 category=Category:Properties of Lie algebras
pageid=2784704 category=Category:Precession
pageid=34281086 category=Category:Spacecraft attitude control
pageid=71398341 category=Category:Supersymmetric quantum field theory
pageid=3062683 category=Category:Topology of homogeneous spaces
pageid=971666 category=Category:Banach algebras
pageid=32862735 category=Category:Bialgebras
pageid=50604436 category=Category:Composition algebras
pageid=14688386 category=Category:Diagram algebras
pageid=3979846 category=Category:Hopf algebras
pageid=39463033 category=Category:Non-associative algebras
pageid=22974471 category=Category:Operator algebras
pageid=2797309 category=Category:Quaternions
pageid=971791 category=Category:Automorphic forms
pageid=962541 category=Category:Fourier analysis
pageid=18672893 category=Category:Singular integrals
pageid=39316758 category=Category:Coding theorists
pageid=2123467 category=Category:Character encoding
```

Logik meiner Akquisition

1. Gewählte Mutter-Kategorie: Category:Mathematics
2. Benutzt offene Wikipedia-API mit **Breadth-First Search (BFS)** um Unterkategorien-Kandidaten zu finden
3. Inspeziere manuell
4. Definitiere ca **800 kewords** manuell (“algebra”, “group theory”, “mathematician”)
5. Filtere Metadata nach keywords
6. Iteriere die letzten 2 Punkte

Weitere Projektziele

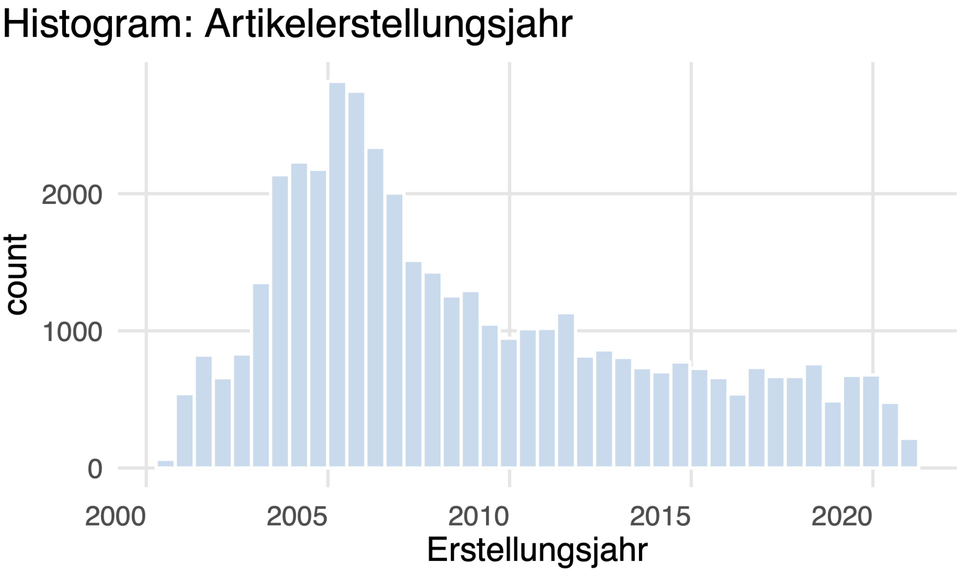
- **Weniger feinemaschige Klassifikation von Texten**
- Erweiterung zu Physik-Artikel
- larger mode: Ließt 6k Wörter
  - Runtime tradeoff?
- Integration in mein Notizbuch-Programm
- Erweiterte Idee: Automatische LLM prompt Generierung mit Mathe-Kategorien-Kontext
  - Design von Quality-Markers für die Artikel? Unterkategorien?

Weitere Projektziele

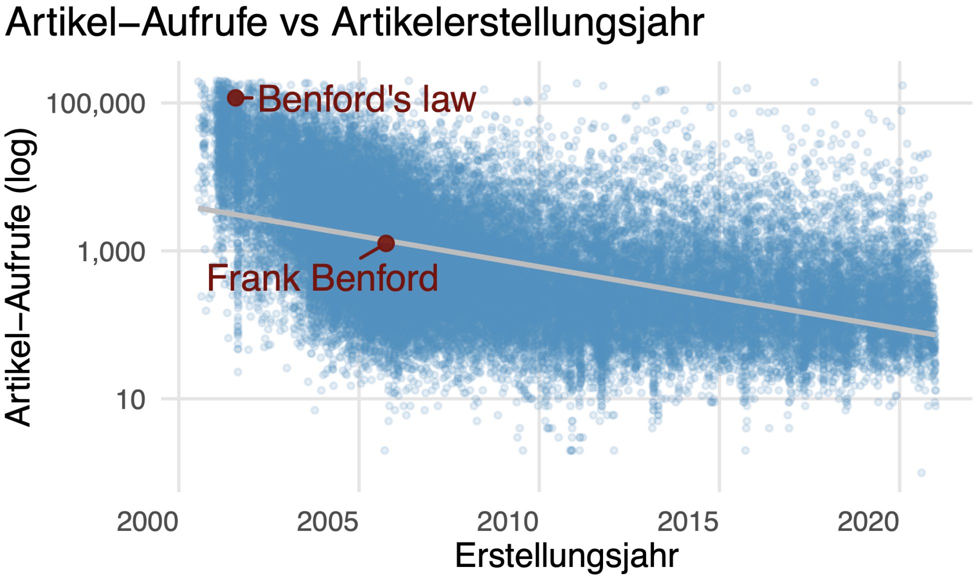
- Weniger feinemaschige Klassifikation

Für meine weiteren Ziehle habe ich bereits zwei Duzend Überkategorien definiert ("Zahlentheorie", "Geometrie", "Statistik", etc.)

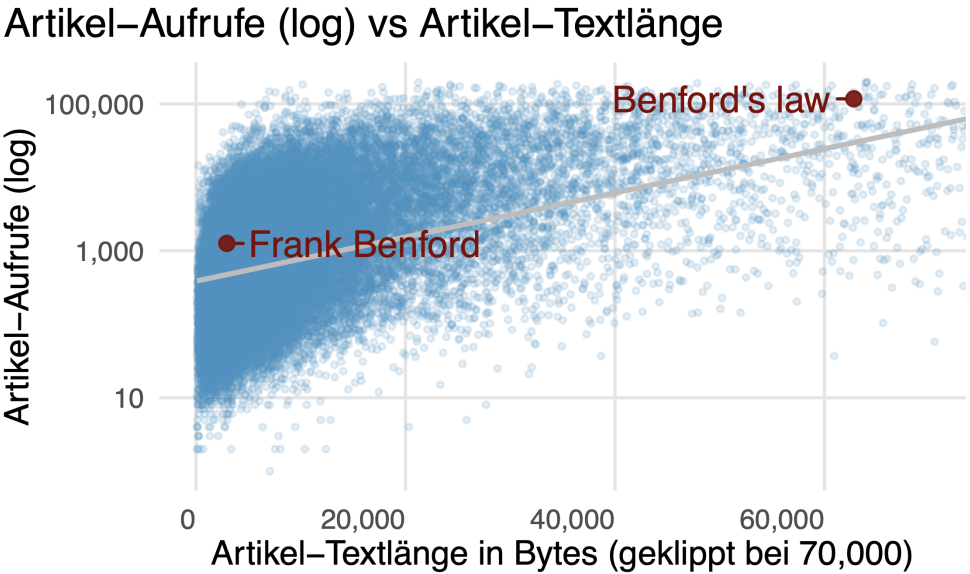
Den Mathe-Datensatz zu verstehen



Den Mathe-Datensatz zu verstehen



Den Mathe-Datensatz zu verstehen



Den Mathe-Datensatz zu verstehen - Statistik

Table 2: 1,607 pages der von mir Klassifizierten Artikel in probability_stats

n_pages	Titel der Wikipedia Kategorie
128	Stochastic_processes
100	Probability_theorems
99	Regression_analysis
97	Probability_stubs
92	Bayesian_statistics
87	Probability_theory
83	Theory_of_probability_distributions
74	Covariance_and_correlation
71	Monte_Carlo_methods
68	Nonparametric_statistics
NA	...
2	Oceanographic_Time-Series
1	Populated_places_in_the_Municipality_of_Markovci
1	WikiProject_Probability_templates

Den Mathe-Datensatz zu verstehen - Statistik

Table 3: Top 7 and bottom 2 pages in der Überkategorie probability_stats

Aufrufe	page	Hauptkategorie
632588	Normal_distribution	Location-scale_family_probability_distributions
528252	Monty_Hall_problem	Probability_theory_paradoxes
406499	Poisson_distribution	Infinitely_divisible_probability_distributions
289053	Exponential_distribution	Infinitely_divisible_probability_distributions
273238	Pearson_correlation_coefficient	Covariance_and_correlation
241534	Gamma_distribution	Infinitely_divisible_probability_distributions
236237	Logistic_regression	Logistic_regression
NA	...	NA
19	Civic_statistics	Applied_statistics
9	Heyde_theorem	Probability_stubs

Den Mathe-Datensatz zu verstehen - Mathematiker

Table 5: Top 10 and bottom 2 pages in der Überkategorie mathematicians

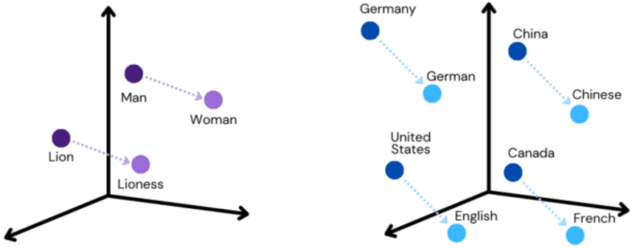
Aufrufe	page	Hauptkategorie
1230395	Ted_Kaczynski	Complex_analysts
1182074	Leonardo_da_Vinci	15th-century_Italian_mathematicians
939498	Alan_Turing	Computability_theorists
670159	Isaac_Newton	History_of_calculus
479642	Socrates	Ancient_Greek_logicians
461580	Aristotle	Ancient_Greek_logicians
444866	Srinivasa_Ramanujan	Indian_combinatorialists
402110	Galileo_Galilei	16th-century_Italian_mathematicians
310478	Confucius	Chinese_logicians
295353	Ada_Lovelace	Ada_(programming_language)
NA	...	NA
3	Simen_Skappel	Norwegian_statisticians
2	Ottaviano_Menni	Italian_mathematician_stubs

Danke!

Fragen?

- Zu Anwendungs-Ideen?
- Zu ChatPGT?

Emergnte Bedeutung



Male – Female

Country – Language

Verb Tense

Den Mathe-Datensatz zu verstehen - Statistik

Table 1: Als Statistik klassifizierte Seiten mit 'Law' im Titel (top 5 und bottom 2 bzgl. Artikel-Aufrufe)

Aufrufe	page	Überkategorie	Hauptkategorie
117088	Benford's_law	probability_stats	Theory_of_probability_distributions
103688	Law_of_large_numbers	probability_stats	Asymptotic_theory_(statistics)
72182	Power_law	probability_stats	Theory_of_probability_distributions
33845	Law_of_total_probability	probability_stats	Probability_theorems
32980	Law_of_total_expectation	probability_stats	Algebra_of_random_variables
NA	...	...	...
524	Blumenthal's_zero–one_law	probability_stats	Probability_theory
149	Engelbert–Schmidt_zero–one_law	probability_stats	Probability_theorems